

BENEFICIAL ROLE OF COMBINED MULTIVITAMIN AND IRON THERAPY AGAINST VITAMIN A IN THE TREATMENT OF IRON DEFICIENT ANEMIC WOMEN

S. ANJUM* AND I.A. RANA

Department of Biochemistry, Hamdard College of Medicine
Hamdard University, Karachi-74600, Pakistan

ABSTRACT:

A review of available information indicates that the global prevalence of anemia, using WHO criteria is around 30%. It is much higher in the developing world, probably around 36% or 1200 million people (Report of a joint FAO/WHO Expert Consultation, 1988).

80 anemic women aged 13-45 y were divided into three groups. They were supplemented with placebo, vitamin A alone, and, multivitamins combined with iron respectively. Maximum benefit in hemoglobin status was achieved in the third group with combined therapy for 8 weeks, while the vitamin A supplemented group failed to show significant improvement in iron status.

Simultaneous therapy with hematenic vitamins and iron resulted in a better response, than, when supplement consisted only of vitamin A. This is in contradiction to many international studies which claim that improvement in vitamin A status results in significant increase in hemoglobin levels (Mohanram et al., 1977; Wolde-Gabriel et al., 1993; and, Suharno et al., 1993).

INTRODUCTION

Nutritional Anemia is a serious problem world wide, and those living in developing countries are at particular risk.

Anemia is defined as a reduction in red cell mass. It is often described as a decrease in the number of red blood cells per cubic mm. or as a decrease in the hemoglobin concentration that is necessary for adequate tissue oxygenation (Keith, 1985).

Anemia is a symptom of many pathological conditions. It is associated with nutritional deficiencies, acute and chronic diseases and may be drug induced. Scattered reports suggest that anemia can be a consequence of vitamins deficiency.

Healthy male subjects fed diet low or deficient in vitamin A but adequate in iron gradually developed anemia which responded to combined vitamin A and iron therapy only (Hodges et al., 1978).

Overt folate deficiency is characterized by presence of megaloblasts in bone

*Correspondence

marrow and blood, and folate deficiency is the second (to iron) most common cause of nutritional anemia (report of a joint FAO/WHO Expert consultation 1988).

Ascorbic acid constantly enhances absorption of non-heme iron from single meals, as indicated by erythrocyte incorporation of isotopic tracer (Halberg et al., 1986 and Consaul et al., 1983).

Iron being an integral component of hemoglobin, myoglobin, and the iron containing enzymes essential for intermediary metabolism, is considered as the main culprit in the development of anemia.

There is sufficient evidence to suggest that iron deficiency is the most significant cause of nutritional anemia in the world. Even by conservative estimates, 600 to 700 million individuals suffer from anemia due to iron deficiency.

Position in Pakistan:

Mushtaq and Jalil (1985) conducted a study to determine the feasibility of fortification best suited to Pakistan, in order to overcome the iron deficiency anemia, and provide a database for the planned 'atta' fortification program. The results of their study are shown in Table-1.

Table 1
Results of the study of Mushtaq and Jalil (1985)
indicating position of anemia in various clusters in Pakistan

Cluster	Iron Deficiency	Folate Deficiency	Vitamin B ₁₂ Deficiency
Pregnant women	83%	25.8%	1.6%
Lactating women	78%	31.6%	-
Adolescent girls	85%	21.9%	-
Adolescent boys	79%	8.7%	-
Children	82.9%	20.9%	0.8%

National Nutrition Survey (1988) indicated that 55% of children in Pakistan are anemic (hemoglobin less than 11%) while 28% are severely anemic. 45% of pregnant and lactating mothers were indicated as being anemic while 10% were found to be severely anemic. 66% of females above the age of 45 years were also found to be anemic.

These observations were based on the recommended allowances and figures that were available before the NNS (1988) on iron. Actual position when/if we analyze this data on the basis of new recommended allowances might be much worse than 1988 survey.

More recently a study was carried out by Imrana Khan (M. Phil. Thesis, 1992) on anemic children from low income groups in Karachi, on the effects of vitamin A supplementation on iron metabolism. It was found that a periodic massive dose of vitamin A may play a key role in improving vitamin A status; however, this investigator found no significant effect of vitamin A supplementation on iron status of these children. There was significant change observed in hematocrit values only.

In the study of Mushtaq and Jalil, the supplementation of vitamin C was not done along with other vitamins. In the study conducted by Imrana Khan, no vitamins other than vitamin A were included. The present study was therefore planned to evaluate the effect of iron and these vitamins (A, B12, C, Folate) on iron status of anemic women.

MATERIALS AND METHODS

The study population comprised of 80 women from Karachi, belonging to lower and lower middle strata of general public. At the beginning of the supplementation trial the women under went clinical and anthropometric evaluations. The clinical characteristics, weight, height and personal data pertaining to monthly income etc. were recorded on a standardized fashion on a special form (table 2). The females were then randomly assigned to any one of the three planned groups as follows:

1. The control group (C): 25 subjects
2. The vitamin A group (VA): 27 subjects
3. The multivitamin group (MVI): 28 subjects

After the initial blood sampling the individuals were provided with the medication in the form of tablets as follows:

- C = placebo
- VA = Vitamin A tablets 50,000 i.u. 1/day
- MVI = Vitamin A tablets 50,000 i.u. 1/day
Ferrous sulphate tablets 100 mg. 2/day
Folic acid tablets 5 mg. 2/day
Vitamin C tablets 200 mg. 3/day
Vitamin B12 tablets 50 ug. 1/day

Subsequent blood sampling was carried out after 4 weeks of taking the first dose and again after another 4 weeks i.e. 8 weeks of taking the first dose.

RESULTS

The data was entered into a computer (*IBM Compatible*) and was analyzed by statistical software *Quattro Pro*.

Hemoglobin Status:

Maximum benefit was achieved in the MVI group in which levels showed significant improvement ($p < .001$) after 8 weeks of intervention. The proportion of women who were cured of anemia was 4% in group C, 3.5% in group VA and 62.9% in the MVI group (table 3).

Serum Iron:

While no changes from baseline were observed in the controls and retinol Supplemented Group, a significant rise (from $53.93 \pm 5.4 \mu\text{g./dl.}$ to $66.48 \pm 5.21 \mu\text{g./dl.}$, ($P = .05$) in this parameter was observed in the Multi-Vitamin and Iron Supplemented Group, in the present study (Table 4). As judged by Serum Iron, 28% women were iron deficient at baseline in the Multi-Vitamin and Iron Supplemented Group. This percentage fell to 21.4% after 4 weeks of intervention indicating improvement, and further fell to 14.8% after 8 weeks. However, we failed to show a significant rise in this parameter, with Retinol supplementation only which is not in line with many studies conducted previously.

Serum Ferritin:

Ferritin is considered the "Gold Standard" for the diagnosis of iron deficiency anemia in pregnancy (Institute of Medicine, Committee on Nutritional Status During Pregnancy and Lactation, 1990; Puolakka et al., 1980).

Since 1922, numerous investigators have shown that anemia may be associated with Vitamin A deficiency (Findlay & Mackenzie, 1922; O' Tool et al., 1974; Hodges et al., 1975). In a 2 year study, conducted on 1-5 years old Guatemalan children, where Vitamin A intake was increased through Vitamin A fortification of sugar, Serum Iron was significantly elevated after 6 months, 1.5, and 2 years of supplementation, whereas Serum Ferritin fell after 6 months but was significantly increased after 1.5 and 2 years (Meija et al., 1982). Our study covered a span of only 8 weeks and did not show any effects on Serum Ferritin levels of Vitamin A supplementation only.

Serum Retinol

When this parameter was evaluated, it was seen that 18% of all (80 women) fell

below the lowest reference limit and only 3.7% remained Retinol deficient after completion of intervention. In group wise analysis, it was observed that, although, the Control Group did not show any improvement, the remaining two groups showed similar improvements. In the Retinol Supplemented Group, mean values showed non-significant improvement ($P = .06$) from baseline after 8 weeks of supplementation, while, in the Multi- vitamin and Iron Supplemented Group, very little improvement ($P = .15$) was seen (Table 5).

RBC Indices:

In the present study, mean Hematocrit values at baseline were sub-normal in all the three groups. After intervention for 8 weeks, significant rise ($P < .01$) was seen from baseline in Retinol Supplemented Group and Multi-vitamin and Iron Supplemented Group ($P < .0001$). In the latter, a significant at 4 weeks of intervention ($P < .01$) (Table 4 & 5).

Vitamin C

It is unclear whether Vitamin C has a direct role in hematopoiesis or if the anemia observed in subjects with Vitamin C deficiency (Scurvy) is a result of the interactions of Ascorbic Acid with Folate and iron metabolism.

In our study subjects, all 80 women had adequate Vitamin C levels but the mean values in all the three groups at baseline were on the lower side of the range. Highly significant improvement ($P < .0001$) was seen only in the Multi-vitamin and Iron Supplemented Group, an expected outcome since this group received Vitamin C supplementation (Table 4). Whether this improved Vitamin C status contributed to the improvement of iron status or not, needs speculation.

Serum Iron values were within normal range in the present study, at baseline, as well as after 4 and 8 weeks in all the three experimental groups. However, this does not indicate that the body iron stores were also adequate in all the subjects. The fact that supplementation with iron as well as Vitamin C resulted in a significant improvement in blood Hemoglobin level, in the third experimental group, clearly suggests that in these subjects, iron supplementation was instrumental in stimulating and helping new Hemoglobin formation, and that Vitamin C probably played a significant positive role in the absorption of the iron supplements and their subsequent utilization for Hemoglobin formation in this group.

Vitamin B₁₂

In all, 17.5% (14/80) women were seen to be deficient in Vitamin B₁₂ status as judged by their serum levels. After intervention for 8 weeks 15% women remained Vitamin B₁₂ deficient in the total samples. In group wise analysis, improvement was

seen in the Multi-vitamin and Iron Supplemented Group where the mean levels improved from 287.07 pg./ml. at baseline to 301.70 pg./ml. after 8 weeks of intervention (Table 4), but this failed to reach the level of significance. The other two groups failed to show any improvement.

Serum Folate

We observed, in our study, that, 11.25% of all the subjects had deficient Folate levels at baseline. After 8 weeks of intervention (5 mg. twice daily) only 5% women remained Folate deficient, all of whom belonged either to controls or Retinol Supplemented Group while all Folate deficient women in the Multi- vitamin and Iron Supplemented Group had reached normally (Table 4).

Serum Vitamin B₁₂ levels were within normal limits in all except 5/25 in controls, 4/28 in Retinol Supplemented Group and 5/27 Multi-vitamin and Iron Supplemented Group. Similarly Folate deficiency was seen in 2/25, 3/28 and 4/27 individuals in the three groups respectively. After supplementation with Vitamin B₁₂ and Folate in the third group the levels of Vitamin B₁₂ and Folate were normalized in all the experimental subjects (except that 4/27 remained Vitamin B₁₂ deficient in the end). In the other two groups the serum Vitamin B₁₂ and Folate values did not show improvements after intervention and those deficient at baseline remained the same after both intervention periods. Furthermore, the MCV remained normal throughout the study. It, therefore, appears that Vitamin B₁₂ and Folate deficiency was not a critical factor in initiating erythropoiesis in the subjects of this study.

TIBC, UIBC and Serum Transferrin

In our study, the mean values for TIBC, UIBC and Serum Transferrin were adequate in all the groups at all the times of intervention. It was seen, however that all three of these parameters correlated well with serum iron values and their adequacy increased with increasing iron concentration with iron supplementation in the Multi-vitamin and Iron Supplemented Group (Table 4).

Total Proteins and A:G Ratio

Since all minerals and vitamins require specific carrier proteins for their transport and storage, a deficiency of proteins may therefore result in anemia. Many previous studies indicate that protein status and presence of protein-energy malnutrition is an important factor in addition to mineral and vitamin deficiencies in the development of anemia (Arroyave et al., 1961; Friend et al., 1961; Oomen, 1958; Teng, 1968).

Ideally, our supplementation program should have included protein supplementation in addition to iron and vitamins, but it was not possible to further complicate an already lengthy study project. Evaluations was carried out however, for proteins, so that actual status of the anemic subjects could be better understood.

It was seen from the results that the mean values in all the three groups at baseline were within normal limits. No changes were observed in any of the groups during or at the end of intervention.

DISCUSSION

The present study was a placebo controlled field trial of anemic women in reproductive age group.

In this study the increase in blood Hemoglobin was significant in the Multi-Vitamin and Iron Supplemented Group and this eliminated anemia in 63% women.

In our study, there were no significant changes in TIBC (which is an indicator of Transferrin), iron or Ferritin in the Retinol Supplemented Group. Significant increase in the Multi-vitamin and Iron Supplemented Group occurred in TIBC. The elucidation of the mechanisms involved will probably need to involve animal studies.

The second question is related to intervention strategies in the long term. Our study has shown that supplementation can virtually eliminate nutritional anemia, but is it feasible through the use of locally available and acceptable foods to achieve the same results?

In Sweden, in 1965, between 25 and 30% of women of child bearing age were anemic. Ten years later the figure had fallen to 6-7% (Halberg et al., 1979). During this period, increased use of oral contraceptives probably decreased menstrual blood loss. The fortifications of all types of flour was increased from 30 to 75 mg. iron/KG. Sales of iron preparations increased and reached over 17000 Kg., equivalent to 5.8 mg. iron/person/day and more than 5 times the amount in United Kingdom. Sales of Ascorbic Acid increase markedly. The authors (Halberg et al., 1979) discussed the relative effects of these changes and concluded that, increased use of medicinal iron accounted for a reduction of 10% in the prevalence of anemia, the higher rate of fortification of flour for a reduction of 7 to 8%.

Keeping in mind the outcome of the above study and the present state of public health education of the vulnerable groups in our country, the food fortification approach, is probably the only suitable target in the long term. For people with very low income, fortification with nutrients during the periods of increased need is perhaps the only feasible option available.

Fortification programs normally supply iron and Folate supplements to women in

the vulnerable age groups (adolescent, pregnant and lactating women), while they do not provide other vitamin supplements. Since the dietary iron in the low income groups is non-heme iron, in the absorption of which Vitamin C has been reported to play a major role (Halberg et al., 1986; Consul & Lee, 1983; Cook & Mansen, 1977), it is therefore recommended that these fortification programs should provide Vitamin C supplements as well, to ensure more absorption of the non-heme iron. Furthermore, since the recommended allowances for iron have been very significantly upgraded by the WHO Expert Group in 1988 as compared to 1973 recommendations of the same group, it is therefore proposed that supplements should be provided at least at the level suggested in the latest WHO Expert Committee Report (1988). The Serum Total Proteins were low in all the groups at 0 and 4 weeks period. They were on the lower limit of the normal in the Control Group and below the minimum normal (6.0 g/dl.) in the Retinol and Multi-Vitamin and Iron Supplemented Group, it is recommended that protein supplements should also be provided to women in the reproductive age group.

AGENDA FOR FUTURE RESEARCH

Part of the problem is that in developing countries, female health care has been a low priority. More research is needed to identify the relative benefits and costs of interventions targeting adolescent and post adolescent women.

It is also important to address the question of whether nutritional interventions in the pre-reproductive period would have a greater impact on reproductive health and reproductive success than interventions during pregnancy alone.

Moreover, there are inadequate data to determine whether giving iron supplements only to women with documented iron deficiency is less, or more cost-effective than routine supplementation. The answer depends on the proportion of women who will require supplements after testing and the costs of the tablets and test.

Studies are also needed to compare the relative effectiveness and cost of routine supplementation and selective supplementation after hematologic and/or iron-storage screening.

If the effectiveness of iron-multi-vitamin supplementation in the reproductive era should be demonstrated, careful thought would have to be given to a program of implementation.

The effects of fortification with iron, Folate and Vitamin C on the Hemoglobin status of women in this group should also be studied. Furthermore, the studies by Mushtaq and Jalil (1985), Khan (1992) and the present study have been carried out for relatively short periods. It is recommended that this fortification strategy should be

evaluated in adolescent and post adolescent age groups in women in this country for longer periods of six months to two years and, the follow up of these cases after the completion of studies should also be carried out to see the long term benefits of these programs.

The latest national nutrition report (1988) shows that anemia is also a major public health problem in women over the age of 45 years. It has been shown by this survey that 66% of women in this age group are anemic and 28% are severely anemic. It is therefore proposed that studies should also be conducted on women belonging to this age group (over 45 years) as well, and that these women are also likely to benefit from the fortification programs.

It is finally recommended that nutrition education should form a regular and relatively more important component of the public health education programs in this country, at school, college and university level as well as through the information media i.e. radio, TV and newspapers etc.

Table 2
Baseline comparison of personal data of the control and supplemented groups

Values are reported as mean and standard error of mean.

Groups	Age (Years)	Height (Feet)	Weight (Kg.)	Blood Pressure (mm Hg)		Monthly Income (RUPEES)	BMI (Wt./Ht ²)
CONTROL (n=25)							
Mean	28.72	5.20	52.98	116.00	67.40	3392.00	21.06
Standard Error	1.91	0.03	1.24	3.95	3.92	243.72	
RETINOL SUPPLEMENTED (n=28)							
Mean	27.25	5.20	51.43	114.21	66.79	3132.14	20.48
Standard Error	1.35	0.03	1.06	4.02	3.56	213.81	
MULTI-VITAMIN & IRON SUPPLEMENTED (n=27)							
Mean	25.59	5.27	52.93	113.81	63.33	3140.74	20.52
Standard Error	1.33	0.03	0.98	3.22	2.25	280.83	

($P > 0.05$ in all parameters i.e. no significant difference between controls and supplemented groups at baseline).

Table 3
Inter group comparison of the mean values after 8 weeks of intervention
Values are reported as mean and standard error of mean.

No.	PARAMETERS	UNITS	MEAN $\pm$ SE OF Groups at 8 week			P-VALUE		
			Mean $\pm$ SE Control	Mean $\pm$ SE Retinol	Mean $\pm$ SE Multi vitamin & Iron	C vs A	C vs M	A vs M
1	Hemoglobin	g./dl.	9.06 $\pm$ 0.29	9.21 $\pm$ 0.16	11.04 $\pm$ 0.29	0.33	0.00	0.00
2	Hematocrit	%	32.65 $\pm$ 1.08	31.45 $\pm$ 0.46	36.73 $\pm$ 0.75	0.16	0.00	0.00
3	RBC Count	$\times 10^6$ /ml.	3.84 $\pm$ 0.13	3.74 $\pm$ 0.08	4.14 $\pm$ 0.11	0.25	0.00	0.05
4	MCV	fl	85.93 $\pm$ 2.34	84.87 $\pm$ 1.66	89.48 $\pm$ 1.42	0.36	0.02	0.10
5	MCH	pg.	24.05 $\pm$ 0.88	24.78 $\pm$ 0.40	26.88 $\pm$ 0.59	0.23	0.00	0.01
6	MCHC	g./dl.	27.90 $\pm$ 0.52	29.32 $\pm$ 0.38	29.99 $\pm$ 0.33	0.02	0.10	0.00
7	Serum Iron	μ g./dl.	54.88 $\pm$ 4.89	52.89 $\pm$ 3.90	66.48 $\pm$ 5.21	0.38	0.02	0.06
8	Serum Ferritin	ng./ml.	18.58 $\pm$ 2.19	19.24 $\pm$ 2.43	28.91 $\pm$ 2.46	0.42	0.00	0.00
9	Serum Transferrin	ng./dl.	137.20 $\pm$ 12.22	132.23 $\pm$ 9.75	166.20 $\pm$ 13.03	0.38	0.02	0.06
10	TIBC	μ g./dl.	160.80 $\pm$ 14.33	158.68 $\pm$ 11.69	199.44 $\pm$ 15.63	0.38	0.02	0.06
11	UIBC	μ g./dl.	109.76 $\pm$ 9.78	105.79 $\pm$ 7.80	132.96 $\pm$ 10.42	0.38	0.02	0.06
12	Total Proteins	g./dl.	6.11 $\pm$ 0.21	5.93 $\pm$ 0.26	5.31 $\pm$ 0.31	0.29	0.07	0.02
13	AG Ratio	RATIO	1.77 $\pm$ 0.16	1.77 $\pm$ 0.13	1.76 $\pm$ 0.10	0.45	0.48	0.47
14	S. Vitamin 'C'	mg./100 ml	0.88 $\pm$ 0.06	0.86 $\pm$ 0.05	1.35 $\pm$ 0.06	0.42	0.00	0.00
15	S. Vitamin 'B12'	pg./ml.	277.96 $\pm$ 18.33	267.14 $\pm$ 13.92	301.70 $\pm$ 16.67	0.32	0.06	0.17
16	S. Folate	ng./ml.	6.84 $\pm$ 0.83	7.76 $\pm$ 0.79	11.40 $\pm$ 0.78	0.21	0.00	0.00
17	S. Retinol	μ g./dl.	30.44 $\pm$ 1.10	32.87 $\pm$ 1.93	32.60 $\pm$ 2.30	0.14	0.46	0.20

Table 4
Comparison of the mean values at 0, 4 and 8 weeks in the Multi-Vitamin and Iron Supplemented Group
Values are reported as mean and standard error of mean.

No.	PARAMETERS	UNITS	MEAN $\pm$ SE OF Multi-vitamin and Iron Supplemented Group			P-VALUE		
			Mean $\pm$ SE at 0 wk.	Mean $\pm$ SE at 4 wk.	Mean $\pm$ SE at 8 wk.	0 vs 4 weeks	0 vs 8 weeks	4 vs 8 weeks
1	Hemoglobin	g./dl.	8.27 $\pm$ 0.23	9.39 $\pm$ 0.26	11.04 $\pm$ 0.29	0.00	0.00	0.00
2	Hematocrit	%	29.16 $\pm$ 0.85	31.96 $\pm$ 0.67	36.73 $\pm$ 0.75	0.01	0.00	0.00
3	RBC Count	$\times 10^6$ /ml.	3.73 $\pm$ 0.17	3.90 $\pm$ 0.17	4.14 $\pm$ 0.11	0.19	0.02	0.04
4	MCV	fl	80.23 $\pm$ 2.43	82.47 $\pm$ 1.89	89.48 $\pm$ 1.42	0.23	0.00	0.00
5	MCH	pg.	22.83 $\pm$ 0.72	24.23 $\pm$ 0.69	26.88 $\pm$ 0.59	0.08	0.00	0.00
6	MCHC	g./dl.	28.49 $\pm$ 0.39	29.30 $\pm$ 0.38	29.99 $\pm$ 0.33	0.07	0.00	0.09
7	Serum Iron	μ g./dl.	53.93 $\pm$ 5.40	58.78 $\pm$ 5.26	66.48 $\pm$ 5.21	0.26	0.05	0.15
8	Serum Ferritin	ng./ml.	16.33 $\pm$ 1.85	20.12 $\pm$ 2.04	28.91 $\pm$ 2.46	0.09	0.00	0.00
9	Serum Transferrin	ng./dl.	134.81 $\pm$ 13.50	146.94 $\pm$ 13.15	166.20 $\pm$ 13.03	0.26	0.05	0.15
10	TIBC	μ g./dl.	161.78 $\pm$ 16.21	176.33 $\pm$ 15.78	199.44 $\pm$ 15.63	0.26	0.05	0.15
11	UIBC	μ g./dl.	107.85 $\pm$ 10.80	117.56 $\pm$ 10.52	132.96 $\pm$ 10.42	0.26	0.05	0.15
12	Total Proteins	g./dl.	5.09 $\pm$ 0.32	5.23 $\pm$ 0.32	5.31 $\pm$ 0.31	0.38	0.31	0.43
13	AG Ratio	RATIO	1.52 $\pm$ 0.10	1.61 $\pm$ 0.09	1.76 $\pm$ 0.10	0.26	0.05	0.14
14	S. Vitamin 'C'	mg./100 ml.	0.84 $\pm$ 0.09	1.12 $\pm$ 0.08	1.35 $\pm$ 0.06	0.01	0.00	0.02
15	S. Vitamin 'B12'	pg./ml.	275.63 $\pm$ 16.80	287.07 $\pm$ 16.21	301.70 $\pm$ 16.67	0.31	0.14	0.27
16	S. Folate	ng./ml.	8.48 $\pm$ 0.94	9.71 $\pm$ 0.88	11.40 $\pm$ 0.78	0.17	0.01	0.08
17	S. Retinol	μ g./dl.	28.92 $\pm$ 2.56	30.75 $\pm$ 2.37	32.60 $\pm$ 2.30	0.30	0.15	0.29

Table 5
Comparison of the mean values at 0, 4 and 8 weeks in the Retinol
Supplemented Group
Values are reported as mean and standard error of mean

NO.	PARAMETERS	UNITS	MEAN $\pm$ SE OF Retinol Supplemented Group			P-VALUE		
			Mean $\pm$ SE at 0 wk.	Mean $\pm$ SE at 4 wk.	Mean $\pm$ SE at 8 wk.	0 vs 4 weeks	0 vs 8 weeks	4 vs 8 weeks
1	Hemoglobin	g./dl.	8.76 $\pm$ 0.15	8.91 $\pm$ 0.15	9.21 $\pm$ 0.16	0.25	0.02	0.09
2	Hematocrit	%	29.68 $\pm$ 0.51	30.56 $\pm$ 0.53	31.45 $\pm$ 0.46	0.12	0.01	0.10
3	RBC Count	$\times 10^6$ /ml.	3.56 $\pm$ 0.08	3.70 $\pm$ 0.08	3.74 $\pm$ 0.08	0.13	0.06	0.39
4	MCV	fl	84.00 $\pm$ 1.46	83.47 $\pm$ 1.72	84.87 $\pm$ 1.66	0.41	0.35	0.28
5	MCH	pg.	24.82 $\pm$ 0.45	24.34 $\pm$ 0.51	24.78 $\pm$ 0.40	0.24	0.48	0.25
6	MCHC	g./dl.	29.64 $\pm$ 0.48	29.24 $\pm$ 0.43	29.32 $\pm$ 0.38	0.27	0.30	0.44
7	Serum Iron	μ g./dl.	50.68 $\pm$ 3.76	51.82 $\pm$ 4.02	52.89 $\pm$ 3.90	0.42	0.34	0.42
8	Serum Ferritin	ng./ml.	19.00 $\pm$ 2.49	18.84 $\pm$ 2.40	19.24 $\pm$ 2.43	0.48	0.47	0.45
9	Serum Transferrin	ng./dl.	126.70 $\pm$ 9.40	129.55 $\pm$ 10.04	132.23 $\pm$ 9.75	0.42	0.34	0.42
10	TIBC	μ g./dl.	152.04 $\pm$ 11.29	155.46 $\pm$ 12.05	158.68 $\pm$ 11.69	0.42	0.34	0.42
11	UIBC	μ g./dl.	101.36 $\pm$ 7.52	103.64 $\pm$ 8.03	105.79 $\pm$ 7.80	0.42	0.34	0.42
12	Total Proteins	g./dl.	5.76 $\pm$ 0.25	5.76 $\pm$ 0.25	5.93 $\pm$ 0.26	0.49	0.32	0.32
13	AG Ratio	RATIO	1.68 $\pm$ 0.14	1.82 $\pm$ 0.13	1.77 $\pm$ 0.13	0.24	0.32	0.41
14	S. Vitamin 'C'	mg./100 ml.	0.74 $\pm$ 0.06	0.75 $\pm$ 0.05	0.86 $\pm$ 0.05	0.45	0.08	0.06
15	S. Vitamin 'B12'	pg./ml.	266.07 $\pm$ 14.24	267.04 $\pm$ 13.76	267.14 $\pm$ 13.92	0.48	0.48	0.50
16	S. Folate	ng./ml.	7.69 $\pm$ 0.77	7.58 $\pm$ 0.78	7.76 $\pm$ 0.79	0.46	0.48	0.44
17	S. Retinol	μ g./dl.	28.48 $\pm$ 2.00	30.80 $\pm$ 2.10	32.87 $\pm$ 1.93	0.21	0.06	0.23

REFERENCES

- Arroyave G., Willson D.D., Mendez J., Behar M., Scrimshaw N.S. (1961). *Am. J. Clin. Nutr.* 9: 180.
- Consul J.R., Lee K. (1983). *J. Agric. Food Schem.* 31: 684.
- Cook J.D., Monsen E.R. (1977). *Am. J. Clin. Nutr.* 30: 235.
- FAO/WHO (1988). Report of a joint FAO/WHO expert consultation. Requirements of vitamin A, iron, folate and Vitamin B12. FAO Food and Nutrition Series, no.23.
- Findlay G.M. and Mackenzie R.D. (1922). *J. Pathol.* 25: 402.
- Friend C.J., Heard C.R.C., Platt B.S., Stewart R.J.C. and Turner M.R. (1961). *Br. J. Nutr.* 15: 231.
- Hallberg L., Bengtsson C., Carby L., Lenartsson J., Rossander L., Tibblin E. (1979). *Bull WHO* 57: 947.
- Hallberg L., Brune M., Rossander L. (1986). *Hum. Nutr. Appl. Nutr.* 40A: 97.
- Hodges R.E.H.E., Sauberlich, J.E., Canham L.A., Meija C., Lykke and Wallace, D.L., (1975). *Abstr.* 3307.
- Hodges R.E., Sauberlich H.E., Canham J.E. (1978). *Am.J. Clin. Nutr.* 31: 876.

- Institute of Medicine (1990). Committee on Nutritional Status During Pregnancy and Lactation. Nutrition during pregnancy. Washington, DC: National Academy Press, 272.
- Keitt A.S., (1985). Introduction to the anemias. In: Wayngaarden J.B., Smith, L. H., eds. Cecil's Textbook of Medicine. 17th ed. Philadelphia: W.B. Saunders., 870.
- Khan I. (1992). Can a periodic massive dose of vitamin A improve the iron status in anemic children of an urban slum of Karachi. M.Phil. Thesis.
- Meija L.A., Arroyave G. (1982). *Am. J. Clin. Nutr.* 36: 87.
- Mohanram M., Kulkarni K.A. and Reddy V. (1977). *J. Vitamin. Nutr. Res.* 47: 389.
- Mushtaq A.K., Jalil F. (1985). Nutritional Anemia: Classification and effect of therapeutic trial for proposed fortification program. Department of Paediatrics, King Edward Medical College, Lahore, p.1.
- National Nutritional Survey (1988). Nutrition Division, National Institute of Health, Special Education and Social Welfare, Ministry of Health, Labour and Family Planning (Health Division), Government of Pakistan, Nutrition Review. Vol.47: 4 April, 1989, p.119.
- O'Tool B.A.R., Fradkin J., Warkany J.G., Wilson and Mann G.V. (1974). *J. Nutr.* 104: 1513.
- Oomen H.A.P.C. (1958). Hypovitaminosis A. Fedn. Proc. Fedn. Am. Soc. exp. Biol. 17, Suppl. 2: 103.
- Puolakkka J., Janne O., Vihko R. (1980). *Acta. Obstet. Gynecol. Scand. Suppl.* 95: 57.
- Suharno D., West C.E., Muhilal, Karyadi D., Joseph G.A.J. (1993). *Lancet.* 342: 1325.
- Teng K.H. (1968). Proc. 1st. natn. Congr. Indonesian Ass. Ophthalmology. p.73.
- Wolde-Gabriel Z., Haile gebru, Tezera Fisseha, West C.E. (1993). *Br. J. Nutr.* 70: 593.